

# L03: Introduction to Keras and Tensorflow

CSci 560 Deep Learning w/ Python (Chollet) Ch. 3

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Summer 2025

# L03.1 What is TensorFlow

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# What is TensorFlow

- TensorFlow is a Python-based open machine learning platform.
- Primary purpose of TensorFlow is to enable researchers to manipulate mathematical expressions over numerical tensors.
- TensorFlow scales fairly well.
- TensorFlow goes beyond scope of NumPy:
  - Can automatically compute the gradient of any differentiable expression.
  - Can run not only on CPUs but also on GPUs and TPUs.
  - Computation defined in TensorFlow can be easily distributed.
  - TensorFlow programs can be exported to other runtimes.



## L03.2 What is 'Keras'

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# What is Keras

- Keras is a deep learning API for Python, built on top of TensorFlow.
- Through TensorFlow, Keras can run on top of different types of hardware (GPU, TPU or plain CPU).
  - And can be scaled to thousands of machines.
- Keras is a “high-level” API that prioritizes developer time/experience.
- Supports several ranges of “workflows”.

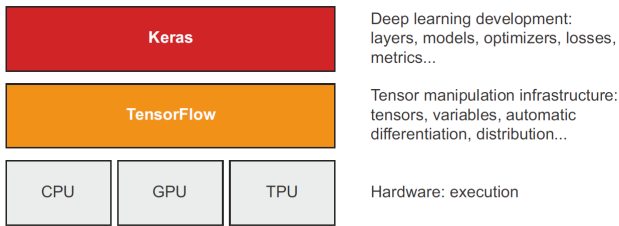


Figure 1: ‘Keras’ and ‘TensorFlow’: ‘TensorFlow’ is a low-level tensor computing platform, and ‘Keras’ is a high-level deep learning AIP. (Chollet, [2021](#), pg.70)

# L03.4 Setting up a Deep Learning Workspace

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# Setting up a Deep Learning Workspace

- To do deep learning on a GPU, you have three options:
  - ① Buy and install a physical NVIDIA GPU on your workstation.
    - [Dettmer The Best GPUs for Deep Learning in 2023](#)
    - [CUDA GPU Compute Capability](#)
  - ② Use GPU instances on Google Cloud or AWS EC2.
  - ③ Use the free GPU runtime from [Google Colab](https://colab.research.google.com/): <https://colab.research.google.com/>

# L03.5 First Steps with TensorFlow

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# L03.6 Anatomy of a Neural Network: Understanding core Keras APIs

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Chollet, F. (2021). *Deep learning with python* (second). Manning.